Laboratory Guide 5: Quantum Portfolio Optimization

Finance & Industry Applications of Variational Quantum Algorithms

Course: Quantum Computing Foundation

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1 Introduction and Objectives

In Lab 4 you built the full technical machinery of variational quantum algorithms: the parameter-shift rule, VQE, and QAOA applied to the MaxCut combinatorial problem. This laboratory is the direct application of that machinery to a real-world industrial problem: **portfolio optimization in quantitative finance**.

The core mathematical structure is identical to Lab 4 — a Quadratic Unconstrained Binary Optimization (QUBO) problem mapped to a QAOA Hamiltonian. The novelty is threefold: (i) the QUBO is derived from real financial data, (ii) interpreting the output requires both a technical and a business lens, and (iii) the problem has a natural classical solution that allows direct benchmarking.

Prerequisites:

- Lab 4: Variational Quantum Algorithms, QAOA, QUBO formulation
- Basic probability and statistics: mean, variance, covariance
- No prior knowledge of quantitative finance is assumed

Learning Objectives:

1. Model a combinatorial portfolio selection problem as a QUBO.
2. Map the QUBO to a quantum Hamiltonian and build the QAOA circuit.
3. Run QAOA on a 4-asset universe (AAPL, MSFT, JPM, XOM) and verify the result against brute-force.
4. Interpret results from both a business and a technical perspective.
5. Understand the scaling advantage for large n and the challenges of NISQ hardware.

Why Quantum for Portfolio Optimization?

For n assets selecting K, the number of candidate portfolios is $\binom{n}{K}$. Classical branch-and-bound solvers handle this well up to $n \approx 1000$. However, adding realistic constraints (transaction costs, sector limits, ESG scores, integer lots) makes the problem NP-hard in general. QAOA provides a *near-term heuristic* that may achieve quantum advantage for $n \gtrsim 100$ constrained assets — a scale that matters for institutional portfolios (e.g. Russell 2000 universe, $n = 2000$).

2 Part A: Classical Portfolio Theory

2.1 Mean-Variance Optimization (Markowitz, 1952)

Let $x \in \mathbb{R}^n$ be a portfolio weight vector with $\sum_i x_i = 1$ (continuous). The mean-variance problem is:

$$\min_x \ \lambda\, x^T \Sigma\, x - \boldsymbol{\mu}^T x, \tag{1}$$

where $\Sigma \in \mathbb{R}^{n\times n}$ is the asset return *covariance matrix*, $\boldsymbol{\mu} \in \mathbb{R}^n$ is the vector of *expected returns*, and $\lambda > 0$ is the *risk-aversion* parameter. The first term penalises risk (portfolio variance), the second rewards expected return.

Risk-Aversion Parameter λ

- $\lambda \to 0$: maximise returns regardless of risk (aggressive investor).
- $\lambda \to \infty$: minimise variance regardless of returns (defensive investor, cash-like).
- Typical institutional values: $\lambda \in [0.5, 5]$.

For this lab we use $\lambda = 1$ as the baseline.

2.2 The Combinatorial Version and its Hardness

In practice, portfolio managers must select *exactly K assets* from a universe of n candidates and invest equally. The binary version of equation (1) is:

$$\min_{x \in \{0,1\}^n} \lambda x^T \Sigma x - \boldsymbol{\mu}^T x, \quad \text{subject to} \quad \mathbf{1}^T x = K. \tag{2}$$

For $n = 4$, $K = 2$ there are only $\binom{4}{2} = 6$ candidates — trivially enumerable. For $n = 50$, $K = 10$ there are $\binom{50}{10} \approx 10^{10}$ candidates. For $n = 500$, $K = 50$: $\sim 10^{82}$ — far beyond brute-force. The continuous relaxation (equation (1)) gives the *efficient frontier* but loses the discreteness that real portfolios require.

2.3 Asset Universe: AAPL, MSFT, JPM, XOM (2020–2024)

We work with the four assets in Table 1. The covariance matrix Σ and expected returns $\boldsymbol{\mu}$ are computed from daily closing prices over 2020–2024 (annualised).

Table 1: Asset data used in this laboratory (annualised, 2020–2024).

Index	Ticker	Sector	μ_i	$\sigma_i = \sqrt{\Sigma_{ii}}$
0	AAPL	Technology	0.340	0.290
1	MSFT	Technology	0.270	0.270
2	JPM	Financials	0.150	0.240
3	XOM	Energy	0.220	0.350

The annualised covariance matrix (approximate):

$$\Sigma = \begin{pmatrix} 0.0841 & 0.0588 & 0.0313 & 0.0203 \\ 0.0588 & 0.0729 & 0.0272 & 0.0170 \\ 0.0313 & 0.0272 & 0.0576 & 0.0294 \\ 0.0203 & 0.0170 & 0.0294 & 0.1225 \end{pmatrix}. \tag{3}$$

The diagonal entries $\Sigma_{ii} = \sigma_i^2$ are the individual asset variances; the off-diagonal entries capture return correlations. AAPL–MSFT have the highest co-movement (both tech-sector), while XOM (energy) is weakly correlated with the others.

3 Part B: QUBO Formulation

3.1 From Constrained to Unconstrained

We convert the equality constraint $\sum_i x_i = K$ into a quadratic penalty added to the objective:

$$\min_{x\in\{0,1\}^n} \underbrace{\lambda\, x^T \Sigma x - \boldsymbol{\mu}^T x}_{\text{cost}} + \underbrace{\lambda_{\text{pen}} \left(\sum_{i=1}^{n} x_i - K \right)^2}_{\text{budget penalty}}, \tag{4}$$

where λ_{pen} must be large enough that any infeasible solution (wrong number of assets) is always worse than any feasible one. A safe choice is $\lambda_{\text{pen}} > \max_{x,\text{feasible}} |C(x)|$.

Expanding and collecting, equation (4) takes the standard QUBO form:

$$\min_{x\in\{0,1\}^n} x^T Q\, x, \tag{5}$$

with:

$$Q_{ij} = \begin{cases} \lambda\, \Sigma_{ii} - \mu_i + \lambda_{\text{pen}}(1-2K) & i = j \\ \lambda\, \Sigma_{ij} + 2\lambda_{\text{pen}} & i \neq j. \end{cases} \tag{6}$$

QUBO Matrix for n

Substituting equation (3) and $\mu = (0.34, 0.27, 0.15, 0.22)$ into equation (6) with $\lambda_{\text{pen}}(1-2K) = 3(1-4) = -9$:

$$Q = \lambda\Sigma - \text{diag}(\boldsymbol{\mu}) + \lambda_{\text{pen}} \begin{pmatrix} 1-2K & 2 & 2 & 2 \\ 2 & 1-2K & 2 & 2 \\ 2 & 2 & 1-2K & 2 \\ 2 & 2 & 2 & 1-2K \end{pmatrix}$$

$$Q = \begin{pmatrix} -8.8259 & 6.1764 & 6.0939 & 6.0609 \\ 6.1764 & -8.9571 & 6.0816 & 6.0510 \\ 6.0939 & 6.0816 & -9.0324 & 6.0882 \\ 6.0609 & 6.0510 & 6.0882 & -8.7575 \end{pmatrix}.$$

The large off-diagonal terms reflect the strong budget penalty; the negative diagonal terms reflect the return incentive.

3.2 Classical Brute-Force Baseline

For $n = 4$, $K = 2$ we can enumerate all $\binom{4}{2} = 6$ portfolios exactly. Table 2 shows the cost $x^T Q x$ and the corresponding financial interpretation.

Table 2: Brute-force enumeration of all 2-asset portfolios ($\lambda = 1$).

Bitstring x	Assets selected	Cost $x^T Q x$	Rank
1100	AAPL + MSFT	−0.335	1 (optimal)
1001	AAPL + XOM	−0.313	2
1010	AAPL + JPM	−0.286	3
0101	MSFT + XOM	−0.261	4
0110	MSFT + JPM	−0.235	5
0011	JPM + XOM	−0.131	6

Financial Interpretation

Why AAPL + MSFT? Despite having the highest inter-asset covariance (both technology sector), AAPL and MSFT also have the highest expected returns (0.34 and 0.27 respectively). For $\lambda = 1$ the return advantage outweighs the diversification cost. As λ increases (more risk-averse investor), the optimal portfolio shifts toward AAPL + XOM (energy provides diversification) and eventually toward JPM + XOM (defensive, low correlation).

4 Part C: Quantum Hamiltonian and QAOA Circuit

4.1 QUBO → Ising Hamiltonian

Substituting $x_i = (1 - Z_i)/2$ into equation (5):

$$\begin{aligned} x^T Q\, x &= \sum_{i,j} Q_{ij} x_i x_j = \sum_{i,j} Q_{ij} \frac{(1 - Z_i)(1 - Z_j)}{4} \\ &= \frac{1}{4} \sum_{i,j} Q_{ij} \big(I - Z_i - Z_j + Z_i Z_j\big) \\ &\equiv H_{\text{portfolio}} \;+\; \text{const.} \end{aligned} \tag{7}$$

Collecting terms, the cost Hamiltonian is:

$$H_{\text{portfolio}} = \sum_{i<j} \frac{Q_{ij}}{2}\, Z_i Z_j + \sum_i \left[-\frac{1}{2} \sum_j Q_{ij} \right] Z_i + \text{const.} \tag{8}$$

Connection to Lab 4

The structure of $H_{\text{portfolio}}$ is a sum of $Z_i Z_j$ and Z_i Pauli operators — *exactly* the same form as the MaxCut Hamiltonian in Lab 4. The QAOA circuit implementation is therefore identical:

- $e^{-i\gamma Z_i Z_j}$: CNOT – $R_z(2\gamma)$ – CNOT
- $e^{-i\gamma Z_i}$: $R_z(2\gamma)$ directly
- $e^{-i\beta H_B}$: $R_x(2\beta)$ on all qubits

Only the coefficients in front of the Pauli strings change.

4.2 Explicit Hamiltonian Terms ($n = 4$, $\lambda = 1$)

Applying equation (8) to our Q matrix yields the following non-trivial Pauli terms (omitting the constant):

Pauli term	Coefficient	Origin
Z_0Z_1	+3.0882	AAPL–MSFT covariance + penalty
Z_0Z_2	+3.0470	AAPL–JPM covariance + penalty
Z_0Z_3	+3.0305	AAPL–XOM covariance + penalty
Z_1Z_2	+3.0408	MSFT–JPM covariance + penalty
Z_1Z_3	+3.0255	MSFT–XOM covariance + penalty
Z_2Z_3	+3.0441	JPM–XOM covariance + penalty
Z_0	+0.1568	AAPL diagonal term
Z_1	+0.0929	MSFT diagonal term
Z_2	−0.0474	JPM diagonal term
Z_3	+0.1238	XOM diagonal term

4.3 QAOA Circuit Structure ($p = 1$)

The $p = 1$ QAOA circuit alternates one cost unitary and one mixer:

$$|\gamma, \beta\rangle = e^{-i\beta \sum_i X_i} e^{-i\gamma H_{\text{portfolio}}} H^{\otimes 4}|0000\rangle. \tag{9}$$

The cost unitary decomposes into one R_z gate per Z_i term and one CNOT$-R_z-$CNOT block per Z_iZ_j term. For $n = 4$: 4 single-qubit + $4 \times 2 = 8$ two-qubit blocks (for $\binom{4}{2} = 6$ edges) = **4 R_z + 6 CNOT pairs** per cost layer. Total circuit depth for $p = 1$: $\mathcal{O}(n^2)$ gates.

5 Part D: Result Interpretation

5.1 Reading the Output Distribution

After QAOA optimisation, we sample the circuit M times and obtain a histogram over $2^n = 16$ bitstrings. The *technical* interpretation: the state $|1100\rangle$ (AAPL=1, MSFT=1, JPM=0, XOM=0, in Qiskit little-endian) should have the highest probability.

Technical → Business Translation

Bitstring	Business meaning
$\|1100\rangle$	Invest in AAPL and MSFT
$\|1001\rangle$	Invest in AAPL and XOM
$\|0011\rangle$	Invest in JPM and XOM
$\|0000\rangle$	Infeasible (no assets)
$\|1111\rangle$	Infeasible (all assets, $K \neq 2$)

Infeasible states have high energy due to the penalty term — they should appear with near-zero probability after optimisation.

5.2 Approximation Quality and Business Risk

QAOA with $p = 1$ does not always find the exact optimum. Define the *success probability* $P^* = P(|1100\rangle)$: the probability the circuit returns the exact optimal portfolio. For $n = 4$, $p = 1$: $P^* \approx 40$–60%. For production use, we would:

1. Sample the circuit M times (e.g. $M = 1000$).
2. Take the most frequent *feasible* bitstring as the answer.

3. Compare its cost against the top-k classical candidates.

Increasing p improves P^* but also increases circuit depth and susceptibility to hardware noise.

5.3 Sensitivity Analysis: Risk Aversion λ

How λ Shifts the Optimal Portfolio

At $\lambda = 0.5$ (return-focused): AAPL + MSFT optimal.
At $\lambda = 1.0$ (balanced): AAPL + MSFT optimal.
At $\lambda = 2.0$ (risk-averse): AAPL + XOM may emerge.
At $\lambda = 5.0$ (very defensive): JPM + XOM (lowest covariance pair) wins.
The quantum circuit must be re-optimised for each λ because the Hamiltonian $H_{\text{portfolio}}(\lambda)$ changes. The optimal QAOA parameters (γ^*, β^*) also shift.

6 Part E: IBM Composer Exercise

Platform: IBM Quantum Composer — `https://quantum.ibm.com/composer`

Student Activity

Visualise the Portfolio Hamiltonian Structure:

1. Build a 4-qubit circuit with H on all qubits.
2. Add the first ZZ term of $H_{\text{portfolio}}$ (AAPL–MSFT, coefficient +3.09): $\text{CNOT}(q_0, q_1)$ — $R_z(2\gamma \cdot 3.09)$ on q_1 — $\text{CNOT}(q_0, q_1)$, with $\gamma = 0.5$.
3. Add the Z_0 single-qubit term (+0.157): $R_z(2\gamma \cdot 0.157)$ on q_0.
4. Apply $R_x(2\beta)$ on all four qubits with $\beta = 0.8$.
5. Measure and observe the Q-Sphere. The states corresponding to feasible 2-asset portfolios (exactly two 1s) should dominate.

Observation: Even one QAOA layer begins to suppress infeasible states. Compare the Q-Sphere with and without the R_z terms.

7 Part F: Qiskit Python Implementation

The full implementation is in the companion notebook. Three exercises are provided:

1. **QUBO Construction:** Build Q from real asset data, enumerate all $\binom{4}{2}$ portfolios classically, identify the optimum.
2. **QAOA Optimisation:** Run QAOA with $p \in \{1, 2\}$, compare the measured distribution against brute-force.
3. **Business Analysis:** Plot the efficient frontier, extract the top-3 most probable portfolios from the QAOA sample, and compute their actual risk-return profiles.

Key Code Structure

```python
# Core objects: same structure as Lab 4, different Hamiltonian
def build_portfolio_hamiltonian(Q: np.ndarray) -> SparsePauliOp:
    n     = Q.shape[0]
    terms = []
    for i in range(n):
        for j in range(i+1, n):
            coeff = Q[i,j] / 2            # ZiZj coefficient
            terms.append((zz_str(i,j,n), coeff))
        z_coeff = -sum(Q[i,:]) / 2        # Zi coefficient
        terms.append((z_str(i,n), z_coeff))
    return SparsePauliOp.from_list(terms).simplify()

# QAOA circuit: identical to Lab 4's build_qaoa_circuit
# Only H_C = H_portfolio changes; circuit structure is the same
```

8 Part G: Scaling and Outlook

8.1 Scaling to Larger Portfolios

Table 3: QAOA circuit scaling for portfolio optimization.

n	$\binom{n}{K}$	QAOA qubits	ZZ terms	Classical gates (p=1)
4	6	4	6	16
6	15	6	15	36
10	252	10	45	100
20	184,756	20	190	400
50	$\sim 10^{12}$	50	1,225	2,500

The QAOA circuit grows as $\mathcal{O}(n^2)$ gates (from the $\binom{n}{2}$ ZZ interaction terms), while the classical search space grows as $\binom{n}{K} \sim n^K/K!$. For $n = 20$, $K = 5$: classical brute-force checks $\sim 10^5$ portfolios (fast classically); for $n = 50$, $K = 10$: $\sim 10^{12}$ (intractable). The QAOA circuit for $n = 50$ requires only 1225 two-qubit gates per layer — within reach of near-term hardware.

8.2 Limitations and Open Questions

- **Barren plateaus:** For $n \gtrsim 50$, the QAOA landscape may be exponentially flat, requiring problem-informed initial parameters.
- **Hardware noise:** The $\sim$1225 CX gates at $n = 50$ with $\sim$0.5% error each give an overall fidelity of $0.995^{1225} \approx 0.002$ — noise dominates.
- **Quantum advantage:** Not yet demonstrated for portfolio optimization. Active research area (JPMorgan, Goldman Sachs, IBM Research).
- **Beyond QUBO:** Transaction costs, lot sizes, turnover constraints are not naturally quadratic — require slack variables or penalty engineering.

9 Part H: Assessment

Q1. QUBO Formulation
Why is the budget constraint $\sum_i x_i = K$ added as a penalty term $\lambda_{\text{pen}}(\sum_i x_i - K)^2$ rather than

enforced directly in the quantum circuit?

A) Because quantum circuits cannot measure the number of 1s in a bitstring.

B) Because the QAOA Hamiltonian must be a sum of Pauli operators (diagonal in the Z-basis), and a soft penalty converts the hard constraint into such a form.

C) Because the penalty term is always zero for feasible solutions, making it costless.

Q2. Hamiltonian Terms

In the portfolio Hamiltonian $H_{\text{portfolio}}$, what is the physical origin of the Z_iZ_j terms?

A) The expected returns μ_i of the individual assets.

B) The covariance Σ_{ij} between assets i and j, plus the budget penalty terms that couple all pairs of qubits.

C) The CNOT gates in the QAOA mixer.

Q3. Result Interpretation

A QAOA circuit for the 4-asset portfolio returns the histogram: $P(|1100\rangle) = 0.45$, $P(|1001\rangle) = 0.22$, $P(|0011\rangle) = 0.15$, all others < 0.05. What is the recommended decision, and why is the second-most-probable state a useful output?

A) Always select $|1100\rangle$; all other states are errors.

B) Select $|1100\rangle$ as primary; $|1001\rangle$ is a useful second-best feasible portfolio that can serve as a diversification alternative.

C) Re-run the circuit until one state reaches $> 99\%$ probability.

Q4. Scaling

For $n = 50$ assets and $K = 10$, the classical brute-force requires $\binom{50}{10} \approx 10^{10}$ cost evaluations. How many two-qubit gates does the QAOA circuit for $n = 50$ require per layer p?

A) $n = 50$

B) $n^2 = 2500$

C) $\binom{n}{2} = 1225$ (one CNOT pair per ZZ interaction term)

Solutions: Q1: B (the Ising Hamiltonian is $\sum_i h_i Z_i + \sum_{ij} J_{ij} Z_i Z_j$; the penalty converts the constraint into this Pauli-operator form); Q2: B (the Q_{ij} for $i \neq j$ contains $\lambda\Sigma_{ij}$ from the risk term plus $2\lambda_{\text{pen}}$ from the penalty, both contributing to the Z_iZ_j coefficient after the substitution $x_i = (1 - Z_i)/2$); Q3: B (in real applications the QAOA output is a probability distribution over solutions; the second-most-probable feasible state is a legitimate portfolio recommendation, not noise — it represents the second-best risk-return trade-off found by the algorithm); Q4: C ($\binom{50}{2} = 1225$ ZZ terms each requiring one CNOT pair, giving $2 \times 1225 = 2450$ CNOT gates per layer, plus 50 R_z gates; total $\mathcal{O}(n^2)$).